Detecting Bipartite structures in near bipartite graphs using motif adjacency matrices

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Abstract

Detecting near bipartite partitions of graphs efficiently is complicated as it is not entirely clear which nodes in the graph should be mis-labelled in order to provide the most meaningful results. In this paper we will propose an organic strategy based on the occurrences of anchored induced subgraphs within a graph to produce a motif adjacency matrix that maximises bipartivity. This motif adjacency matrix (MAM) will then be subject to a partitioning by the Fielder vector to produce a near-bipartite partition of the graph. The strategy is applied to five real world datasets and compared to another previously explored gold standard strategy for detecting near bipartite partitions of graphs. Overall, we find that using MAM's that maximise bipartivity is excellent in scenarios with few intra-community edges. However, the approach is limited by skewing partitions towards community detection when no organically occurring "good" bipartition is present.

Introduction

Graphs are in every element of our everyday lives and are fundamental to the way we understand and catalogue information. The associations between our friends form social networks and the way we consider nutrition in our diets forms a taxonomical web of healthy and unhealthy foods. The way we form these classifications, healthy and unhealthy for example is incredibly important as these abstractions form the basis from which we make decisions. Sets of objects that are grouped together are said to have a community structure if they all share similar properties or have a higher intra-community structure with minimal intercommunity connectivity. Consider the set of breads; all breads have a similar nutritional content, as well as structure, colour and ingredients, all that combined makes them bread. Hence the set of all breads within our foods have some similarities that are easy for us to understand. However, there are some community phenomena that are not so obviously identifiable, such as bipartite structure.

Bipartite graphs are graphs in which each node i can be part of a set S_1 such that each neighbouring vertex j in the graph is a member of set S_2. Hence any walk on a bipartite graph must alternate between nodes from the set's S_1 and S_2. This means there exist two independent sets within the graph, which are communities in an abstract sense. Near bipartite graphs are graphs in which there are "some" edges present between nodes from the set S_1 to other nodes in S_1. Finding near bipartite partitions of graphs is more difficult than strictly bipartite graph as it is not immediately clear which of the sets $\{S_1, S_2\}$ node i belongs to. In this paper we will explore how we can use induced subgraphs called motifs to find near bipartite structures within graphs.

A subgraph is small graph, that we will find occurrences of in a larger graph (consider the number of triangles that occur in a graph as an example). Based on the occurrences

of subgraphs within a graph we can determine a new matrix called the motif adjacency matrix that describes the structural connectivity of nodes belonging to occurrences of that subgraph. This concept of a motif adjacency matrix is incredibly powerful and allows us to turn directed graphs, for which bipartivity cannot be determined into a new symmetric weighted representation for which near bipartite partitions can be determined through traditional community detection.

Methods

Consider the graph $G = (V, E)$ with vertex set $V = \{1, \dots, k\}$ and edge set $E \subseteq (V \times V)$ wherein each element in E is distinct and $(v_i, v_i) \notin E \forall v_i \in V$. We will define the Adjacency matrix, A, of the graph G as:

$$A_{ij} = \begin{cases} 1 & if\ (i,j) \in E \\ 0 & if\ (i,j) \notin E \end{cases},\ \forall 1 \leq i, j, \leq k$$

A structural motif is an induced subgraph, $G_{\mathcal{M}} = (V_{\mathcal{M}}, E_{\mathcal{M}})$ on G, and given $S \subseteq V$ is in $Occ_{\mathcal{M}}(G) \Leftrightarrow G_S = (S, E_S = S \times S \cap E)$ is isomorphic to $G_{\mathcal{M}}$. This formalism gives us the foundation to define a motif adjacency matrix, M, with anchor set $A \subseteq V$ as:

$$M_{ij} = \left|\{S \in Occ_{\mathcal{M}}(G) : \exists \sigma \in I_{\mathcal{M},G}(S) :\ \sigma(i), \sigma(j) \in A\}\right|$$

Where $I_{\mathcal{M},G}(S)$ contains the set of all the isomorphisms between G_S and the motif $\mathcal{M}$ [1]. The motif adjacency matrix (MAM) M is weighted and symmetric and thus has a graph Laplacian L:

$$L = D - M, \text{ where } D_{ii} = \sum_{i=1}^{k} M_{ij}$$

Using L we can determine an optimal partition of the graph G by assigning nodes v_i to the set S_1 where $x_i > 0$ and nodes v_j to set S_2 where $x_j \leq 0$ where x is the eigenvector corresponding to the first non-zero eigenvalue of L:

$$Lx = \lambda x$$

Hence the sets $\{S_1, S_2\}$ that we assign the nodes v_i to will have a near-bipartite community structure if the motif used to generate the MAM $G_{\mathcal{M}}$ has a bipartite structure. To test the quality of near-bipartite partitions generated using this strategy we will consider the modularity function $Mod(A, s)$ [2]:

$$Mod(A, \{S_1, S_2\}) = \frac{1}{2m} \sum_{ij} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \frac{S_i S_j + 1}{2}$$

Where k_i is the expect degree of node v_i and m is the total sum of all edges in A. For a given partition $\{S_1, S_2\}$ modularity is lower bounded at -0.5 if the partition is strictly bipartite and upper bounded at 1 if the partition has a fully connected community structure. We will compare the modularity of partitions using the described method to a previously explored method for detecting near bipartite partitions, which is to take the eigenvector of the adjacency matrix and assign node v_i to the set S_1 where $x_i > 0$ and nodes v_j to set S_2 where $x_j \leq 0$.

We will test this method on 5 different datasets, the graph of British airways flights [3], the social relationships between members of a karate club [4], the structural connections of electrical components, the interaction web of gene within E coli [5] and a protein interaction dataset within yeast [6]. The motifs we will be testing are Q13260+0.1*T78 with anchors $\{1,2\}$ and $\{2,3\}$ respectively as shown in Fig 1. Where vertices in red denote membership to set 1 and black denote membership to set 2.

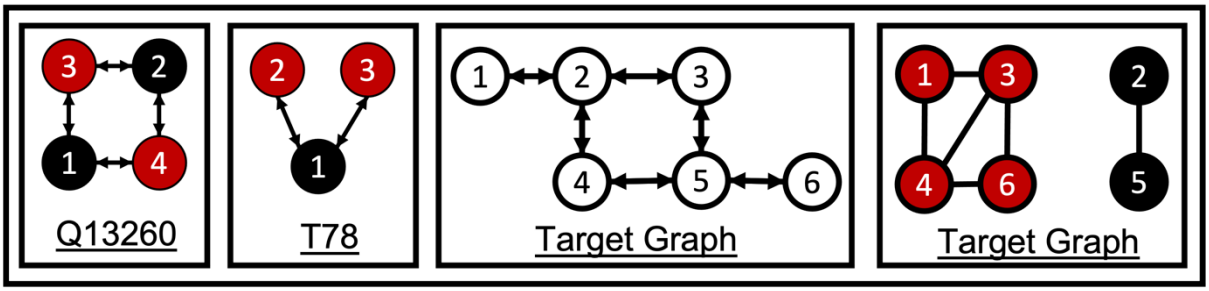


Fig 1. Far left and centre left are the motifs of interest. Centre right is the target graph and far right is the corresponding motif adjacency matrix, clearly separated by colour for the two independent bipartite sets.

Results

The results for both methods applied on the five datasets are shown in the table below:

Name	**Modularity using MAM**	**Modularity using A**
British Airways	-0.4686	-0.4686
Karate Club	-0.163	-0.272
Electrical	0.098	-0.337
E coli	-0.458	-0.473
Yeast	0.030	-0.413

Table 1. Modularity values for the method described in this paper and the previously explored method based on the adjacency matrix.

In some cases, the proposed method matches the modularity produced by the partitioning based on the adjacency matrix. However, in the electrical and yeast datasets, it doesn't even produce a bipartite partitioning at all. In toy examples and strictly bipartite cases, the strategy works flawlessly, so there appears to be an issue from specific intra-community edges that skews the MAM method away from producing bipartitions of non-bipartite datasets. Consider the toy examples in Fig 2. to explain this phenomenon.

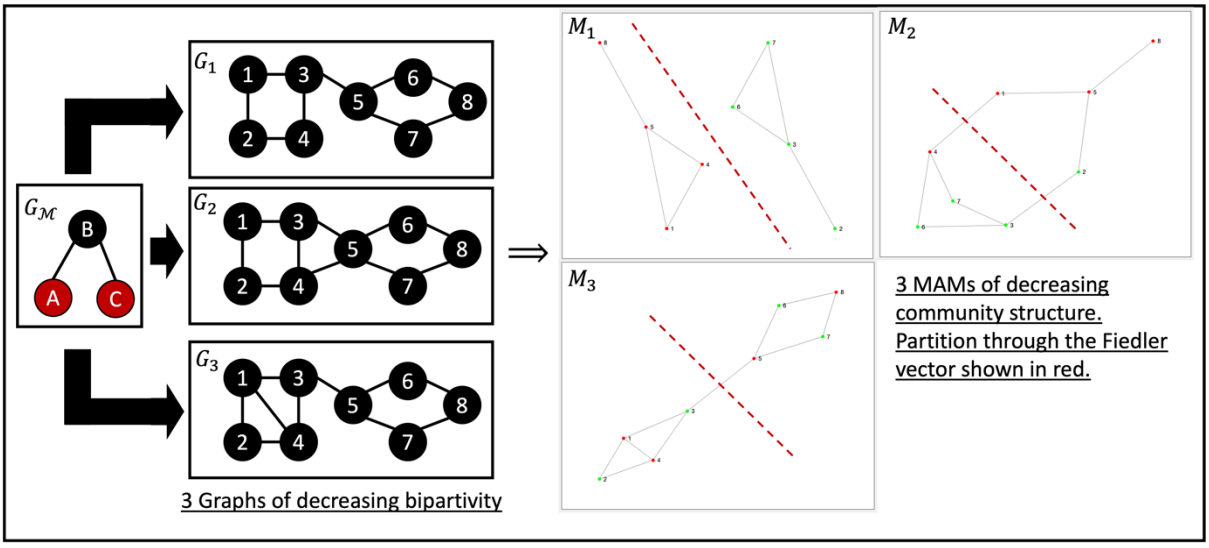


Fig 2. Shows the motif $G_{\mathcal{M}}$ applied to graphs G_1, G_2 and G_3 with corresponding motif adjacency matrices M_1, M_2 and M_3. Partition lines shown in dashed red with true partition colours shown in red and green.

In Figure 2. we see G_1 is a traditional strictly bipartite network wherein each node v_i can be assigned to an independent set S_1 or S_2. In this case our proposed strategy works flawlessly, however this case is somewhat trivial. In the graph G_2 we see the introduction of a single inter-community edge, results in the misclassification of a pair of nodes. This is the case like in the karate club dataset, wherein we find a bipartition, but it has a modularity of closer to 0 because of these inter-community edges. This type of edge is additive, it adds new occurrences of the motif $G_{\mathcal{M}}$ that result in a MAM with a single clustered component rather than two independent ones. Worse than this, however, is the final case in G_3 that has another intracommunity edge that is devastating to the detection of bipartite partitions using our proposed strategy. Because the edge connected in such a way as to remove an occurrence of the motif, it is not only additive but deconstructive, as it removes occurrences that we need whilst adding new occurrences that are not bipartite. This type of phenomena is what occurs during the Electrical and Yeast datasets where the method pivots toward detecting community structure rather than bipartite structures.

Conclusion

Whilst the results shown in this paper are damning for this method's applicability to the detection of solely bipartite partitions from near-bipartite graphs, there is an interesting application in situations wherein you don't know whether your dataset is near bipartite or has community structure. If it has a community structure, this method will tend towards emphasising it, whereas if it has a bipartite structure it will tend towards producing a partition that highlights that structure instead. Whilst this has limited use in a practical sense it is very organic and may have niche uses in dataset wherein the macro-scale structural connectivity is unknown.

Future work

In this paper we have limited our research to only exploring a linear combination of the motifs T78 and Q13260, where there are 10+ potentially useful bidirectional bipartite motifs that could produce meaningful results if combined appropriately. Furthermore, we've limited the datasets we're working on to bidirectional ones, where we may find more success applying this strategy to directed graphs. Because of the definition of the occurrence within $Occ_{\mathcal{M}}(G)$ this analysis is constrained to consider only structural occurrences of the motif $G_{\mathcal{M}}$ in G. Functional occurrences are an alternative approach wherein we don't require a perfect one-to-one mapping of $G_s \rightarrow G_{\mathcal{M}}$ and a linear combination of structural and functional motifs may provide a strategy around the deconstructive edges shown in Fig 2.G_3 albeit at the cost of more additive edges.

References

[1] A. Benson, Tools for higher-order network analysis, Stanford University , 2017.

[2] M. Newman, "Modularity and community structure in networks," *PNAS: Applied Mathematics,* vol. 103, no. 23, pp. 8577-8582, 2006.

[3] A. Cardillo and et al, “Emergence of network features from multiplexity,” *Scientific reports 3,* no. 1344, 2013.

[4] W. Zachary, “An information flow model for conflict and fission in small groups,” *Anthropology research,* no. 33, pp. 452-473, 1977.

[5] K. Nakai and M. Kanehisa, “Expert system for predicting protein localization sites in gram-negative bacteria,” *Proteins,* vol. 2, no. 11, pp. 95-110, 1991.

[6] B. Dongbo, Z. Yi, C. Lun, X. Hong and et al, “Topological structure analysis of the protein-protein interaction network in budding yeast,” *Nucleic acids reserach,* p. 2443–2450, 2003.